

TE Curve Verification Pipeline Familywise ONNX Report - wave4_1_smooth_l1_robust_loss

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml1`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_smooth_l1_robust_loss/2026-07-13-09-08-55__track2_wave4_1_smooth_l1_robust_loss_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_smooth_l1_robust_loss/2026-07-13-09-08-55__track2_wave4_1_smooth_l1_robust_loss_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_smooth_l1_robust_loss/2026-07-13-09-08-55__track2_wave4_1_smooth_l1_robust_loss_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_1_smooth_l1_robust_loss/2026-07-13-09-08-55__track2_wave4_1_smooth_l1_robust_loss_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave4_1_smooth_l1_robust_loss` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave4_1_smooth_l1_robust_loss_fw_simplified_setpoints</code>	<code>2026-07-12-20-38-09__te_wave4_1_smooth_l1_robust_loss_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave4_1_smooth_l1_robust_loss_bw_simplified_setpoints</code>	<code>2026-07-12-21-02-43__te_wave4_1_smooth_l1_robust_loss_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave4_1_smooth_l1_robust_loss_global_simplified_setpoints</code>	<code>2026-07-12-20-13-14__te_wave4_1_smooth_l1_robust_loss_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/forward/2026-07-12-20-38-09__te_wave4_1_smooth_l1_robust_loss_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/forward/2026-07-12-20-38-09__te_wave4_1_smooth_l1_robust_loss_fw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=086-val_mae=0.00353570.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/backward/2026-07-12-21-02-43__te_wave4_1_smooth_l1_robust_loss_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/backward/2026-07-12-21-02-43__te_wave4_1_smooth_l1_robust_loss_bw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=142-val_mae=0.00357827.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/global/2026-07-12-20-13-14__te_wave4_1_smooth_l1_robust_loss_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/global/2026-07-12-20-13-14__te_wave4_1_smooth_l1_robust_loss_global_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=087-val_mae=0.00364010.ckpt</code>

Aggregate Metrics

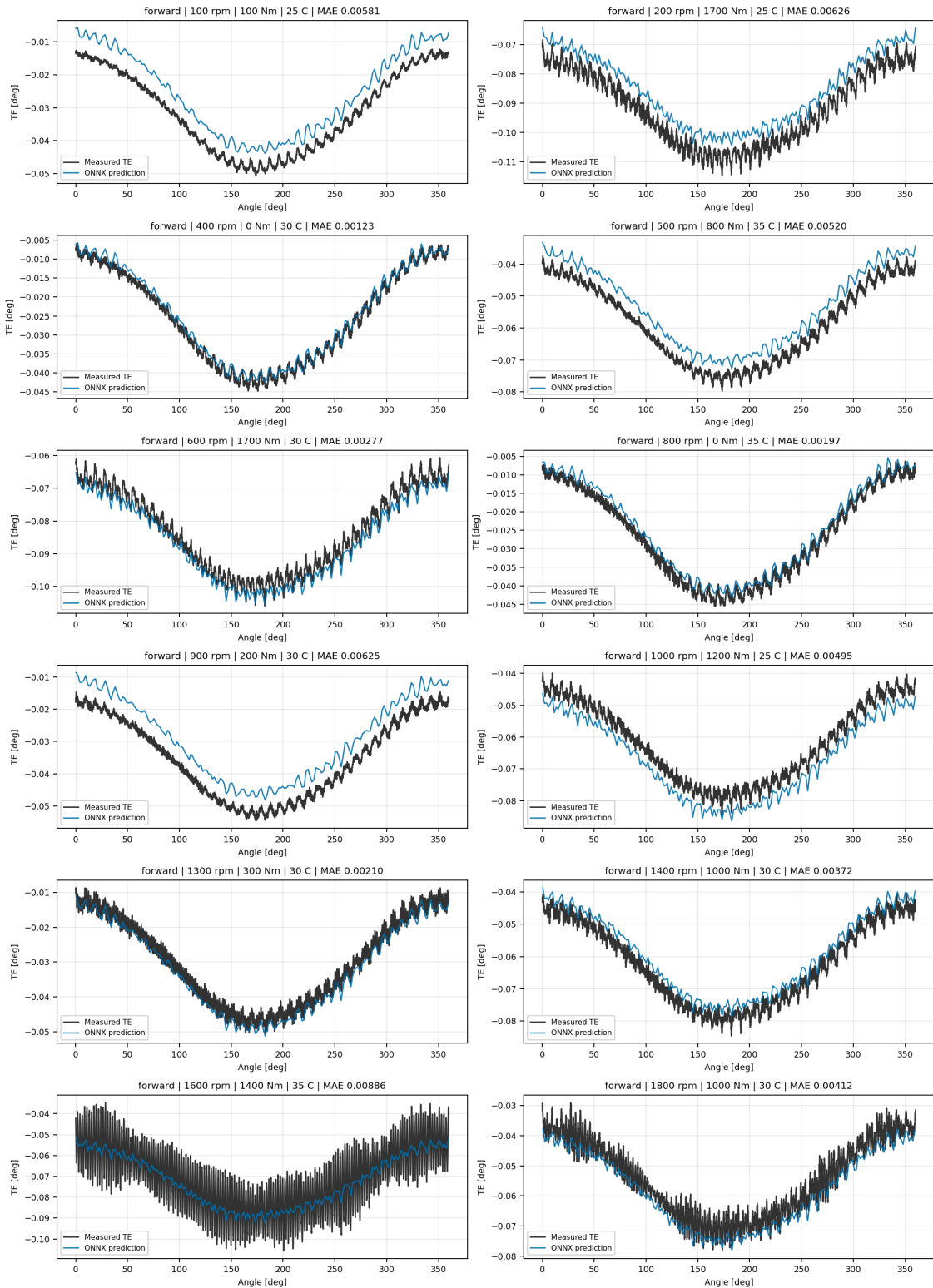
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003305	0.003598	7.750	14.360
backward	97	0.003527	0.003848	8.180	13.736
global	194	0.003373	0.003685	7.861	14.039

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000344	0.002896	0.001326	0.002823
backward	-0.000246	0.002952	0.001533	0.003340
global	0.000554	0.002878	0.001431	0.003093

Forward 12-Curve Page

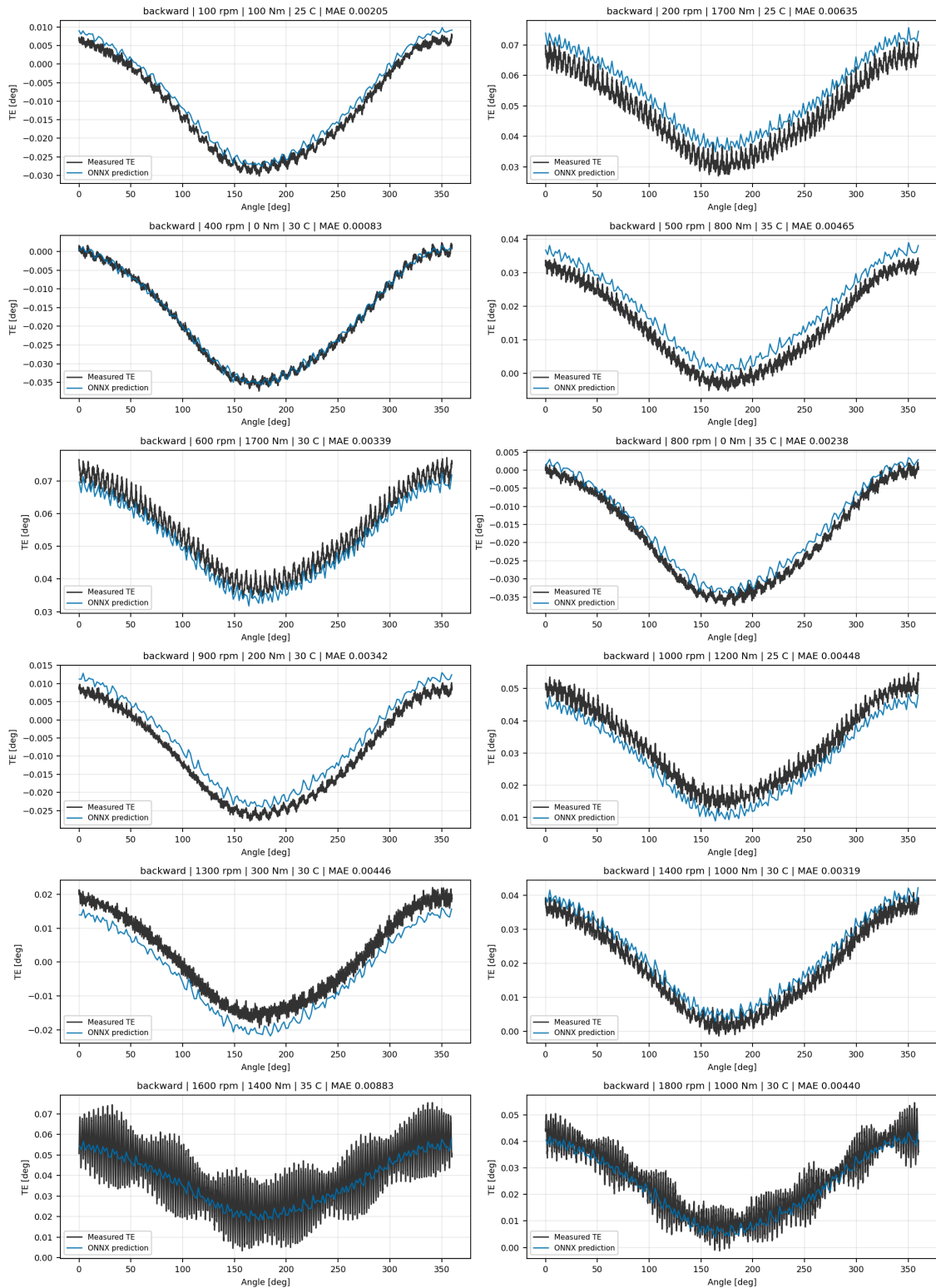
Simplified Dataset + Setpoints | wave4_1_smooth_l1_robust_loss | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

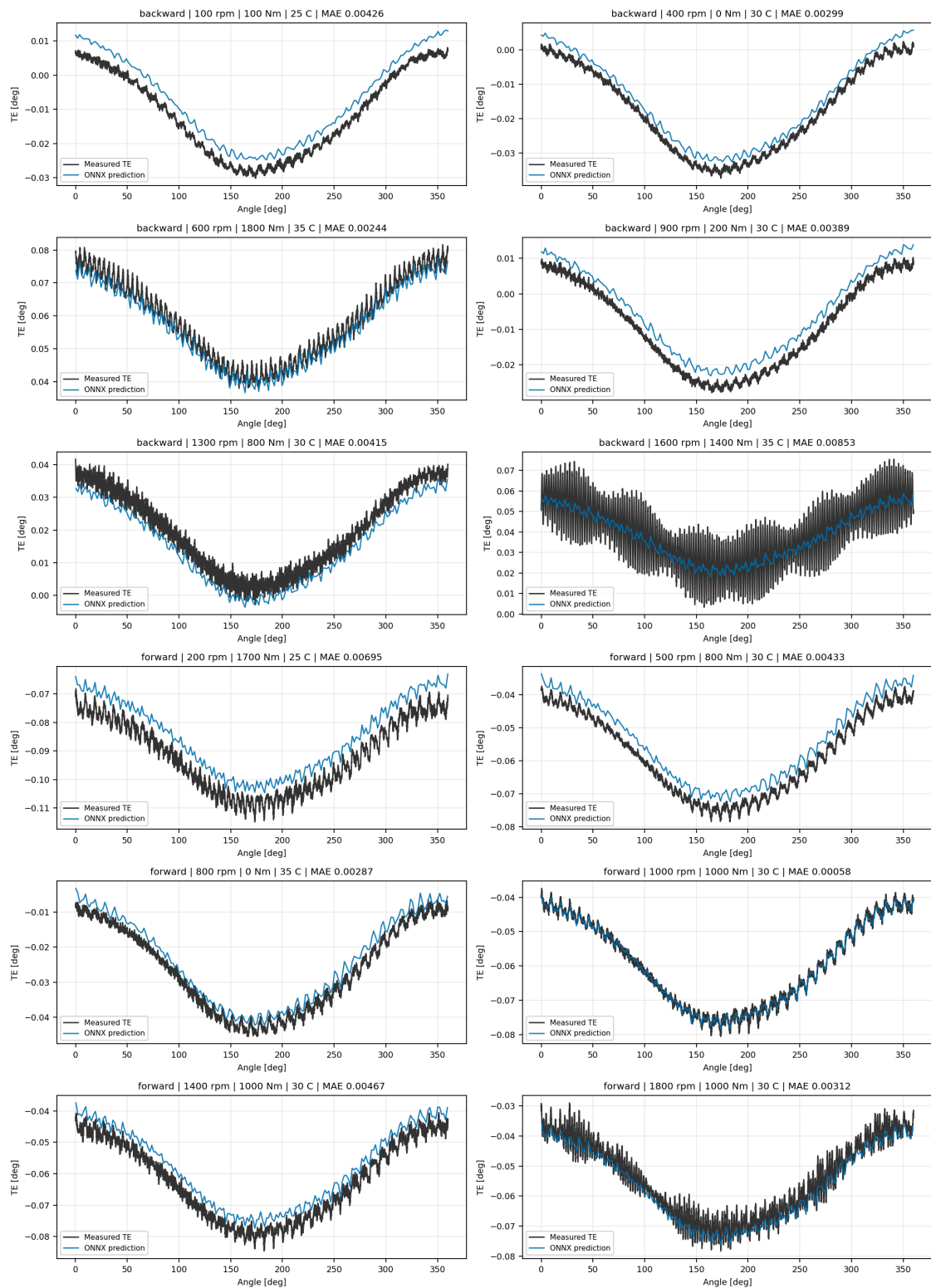
Simplified Dataset + Setpoints | wave4_1_smooth_l1_robust_loss | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave4_1_smooth_l1_robust_loss | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave4_1_smooth_l1_robust_loss ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_1_smooth_l1_robust_loss_fw__polished_setpoints	2026-07-12-23-00-14__te_wave4_1_smooth_l1_robust_loss_fw__polished_setpoints	polished_setpoint_curve_v1
backward	te_wave4_1_smooth_l1_robust_loss_bw__polished_setpoints	2026-07-12-23-26-15__te_wave4_1_smooth_l1_robust_loss_bw__polished_setpoints	polished_setpoint_curve_v1
global	te_wave4_1_smooth_l1_robust_loss_global__polished_setpoints	2026-07-12-22-12-14__te_wave4_1_smooth_l1_robust_loss_global__polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/forward/2026-07-12-23-00-14__te_wave4_1_smooth_l1_robust_loss_fw__polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/forward/2026-07-12-23-00-14__te_wave4_1_smooth_l1_robust_loss_fw__polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=079-val_mae=0.00192933.ckpt
backward	models/polished_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/backward/2026-07-12-23-26-15__te_wave4_1_smooth_l1_robust_loss_bw__polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/backward/2026-07-12-23-26-15__te_wave4_1_smooth_l1_robust_loss_bw__polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=115-val_mae=0.00193797.ckpt
global	models/polished_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/global/2026-07-12-22-12-14__te_wave4_1_smooth_l1_robust_loss_global__polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_1_smooth_l1_robust_loss/global/2026-07-12-22-12-14__te_wave4_1_smooth_l1_robust_loss_global__polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=143-val_mae=0.00190163.ckpt

Aggregate Metrics

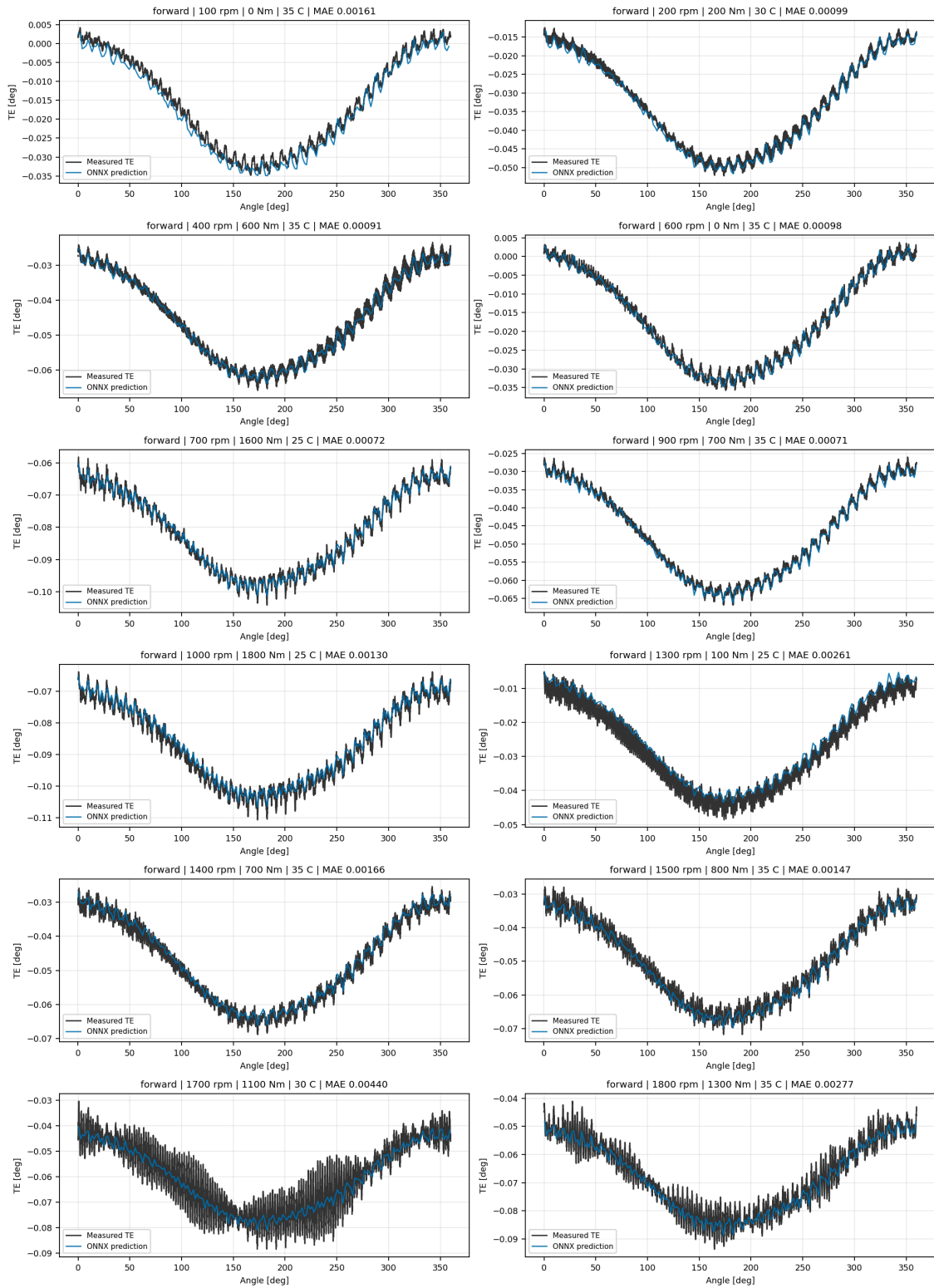
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001947	0.002307	4.279	9.489
backward	94	0.002523	0.002967	4.608	10.754
global	194	0.002182	0.002583	4.338	10.860

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000150	0.001029	0.001532	0.003657
backward	0.000721	0.001053	0.002113	0.005803
global	0.000287	0.000914	0.001828	0.004512

Forward 12-Curve Page

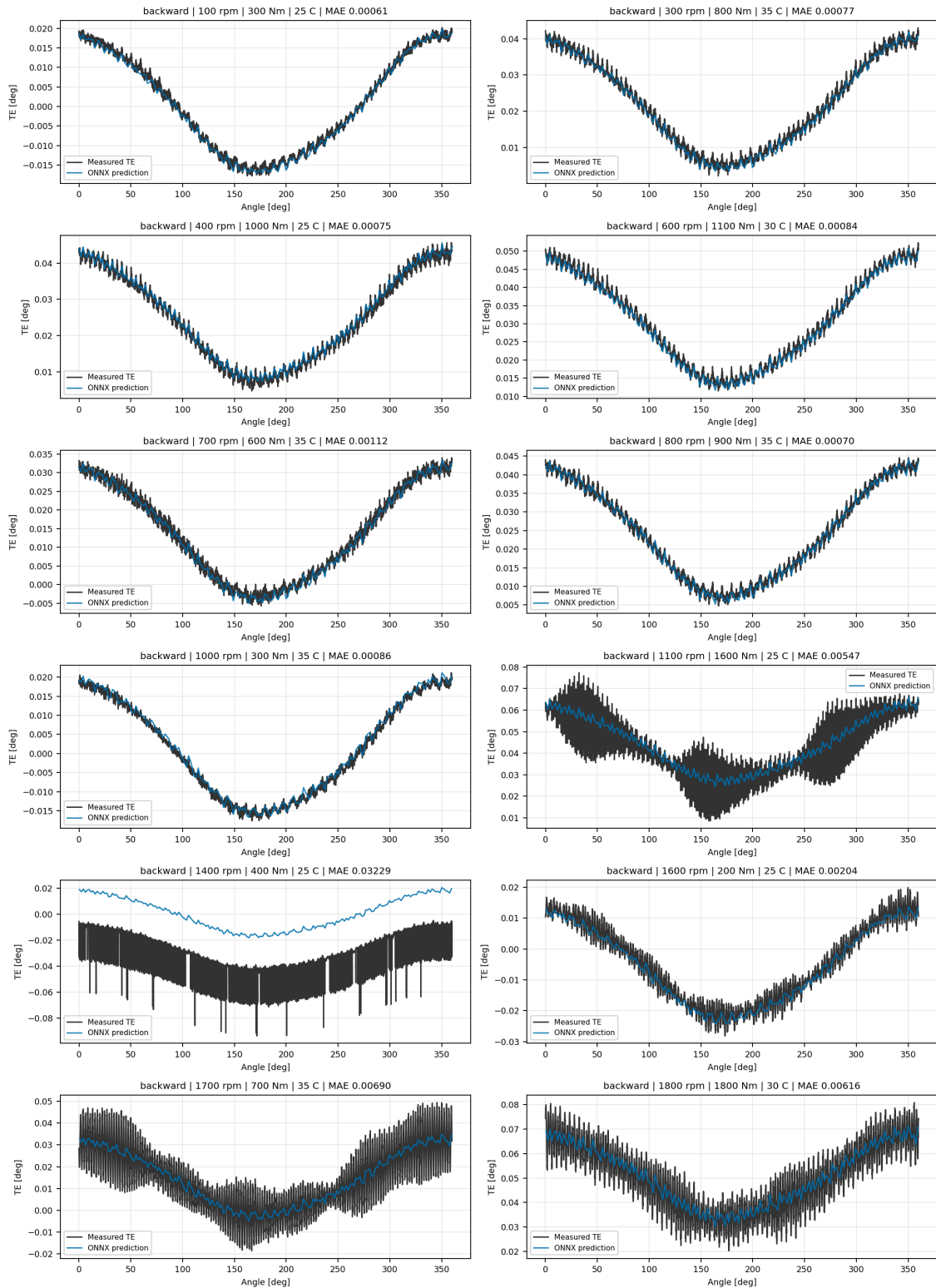
Polished Dataset + Setpoints | wave4_1_smooth_l1_robust_loss | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

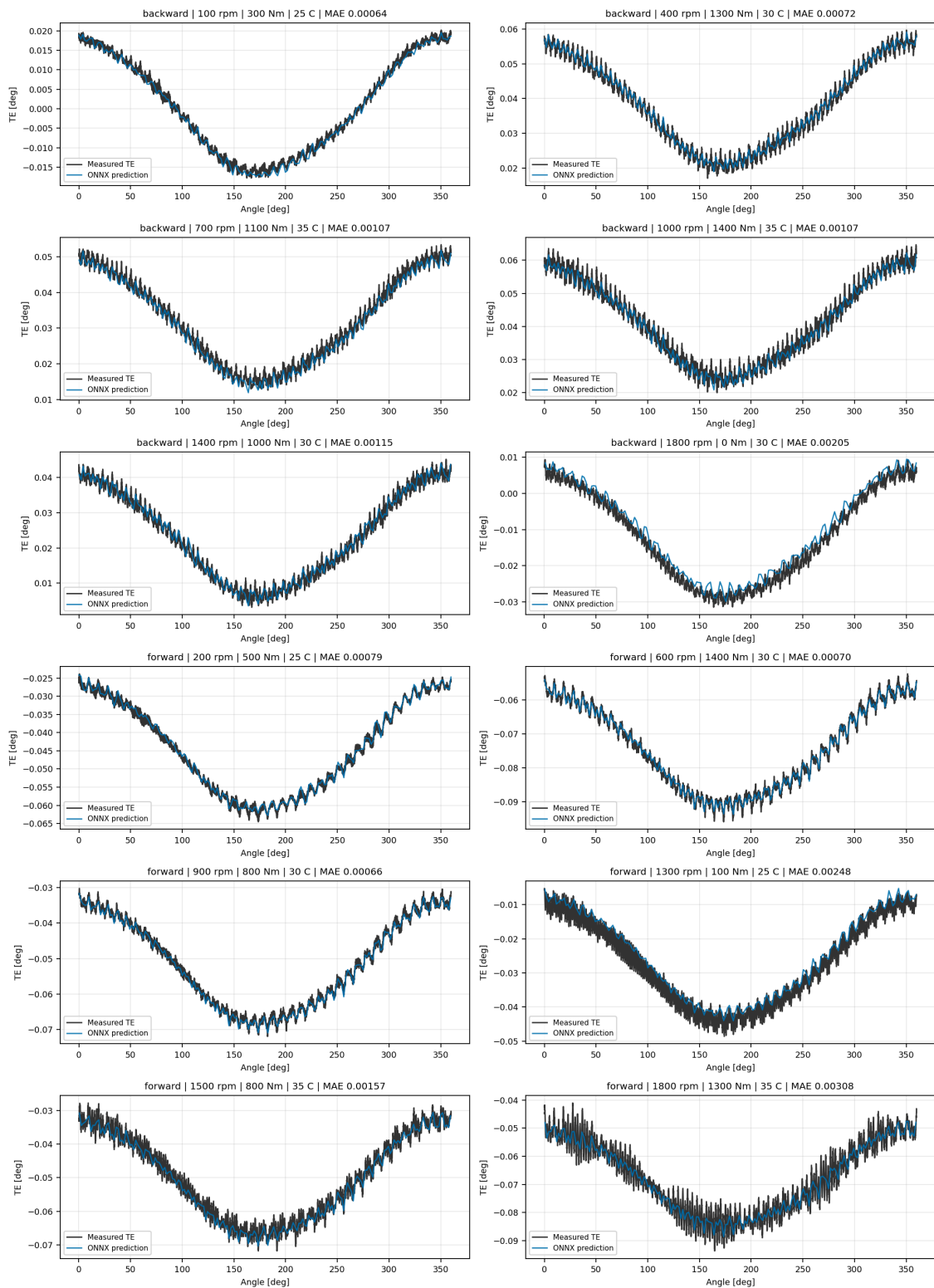
Polished Dataset + Setpoints | wave4_1_smooth_l1_robust_loss | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave4_1_smooth_l1_robust_loss | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave4_1_smooth_l1_robust_loss ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_1_smooth_l1_robust_loss_fw__polished_actual_values	2026-07-13-00-54-21__te_wave4_1_smooth_l1_robust_loss_fw__polished_actual_values	polished_point_v1
backward	te_wave4_1_smooth_l1_robust_loss_bw__polished_actual_values	2026-07-13-01-50-39__te_wave4_1_smooth_l1_robust_loss_bw__polished_actual_values	polished_point_v1
global	te_wave4_1_smooth_l1_robust_loss_global__polished_actual_values	2026-07-13-00-17-37__te_wave4_1_smooth_l1_robust_loss_global__polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave4_1_smooth_l1_robust_loss/forward/2026-07-13-00-54-21__te_wave4_1_smooth_l1_robust_loss_fw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_smooth_l1_robust_loss/forward/2026-07-13-00-54-21__te_wave4_1_smooth_l1_robust_loss_fw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=218-val_mae=0.00183482.ckpt
backward	models/polished_dataset/actual_values/exported/wave4_1_smooth_l1_robust_loss/backward/2026-07-13-01-50-39__te_wave4_1_smooth_l1_robust_loss_bw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_smooth_l1_robust_loss/backward/2026-07-13-01-50-39__te_wave4_1_smooth_l1_robust_loss_bw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=128-val_mae=0.00189374.ckpt
global	models/polished_dataset/actual_values/exported/wave4_1_smooth_l1_robust_loss/global/2026-07-13-00-17-37__te_wave4_1_smooth_l1_robust_loss_global__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_smooth_l1_robust_loss/global/2026-07-13-00-17-37__te_wave4_1_smooth_l1_robust_loss_global__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=151-val_mae=0.00188208.ckpt

Aggregate Metrics

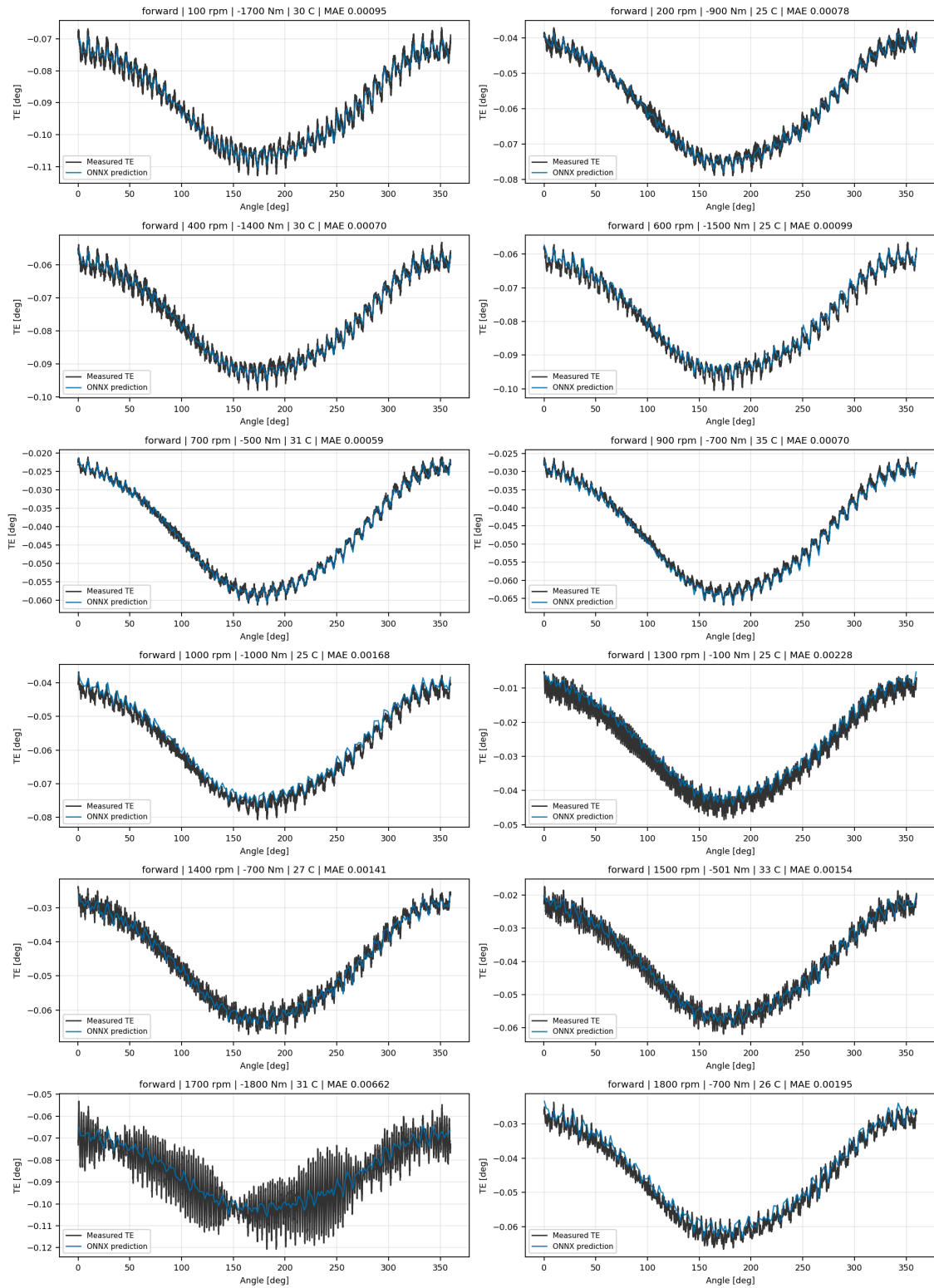
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001758	0.002123	3.825	9.160
backward	94	0.002200	0.002646	4.148	10.462
global	194	0.002028	0.002438	4.123	10.298

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000064	0.000705	0.001555	0.003632
backward	0.000327	0.000600	0.002094	0.005637
global	0.000040	0.000785	0.001819	0.004571

Forward 12-Curve Page

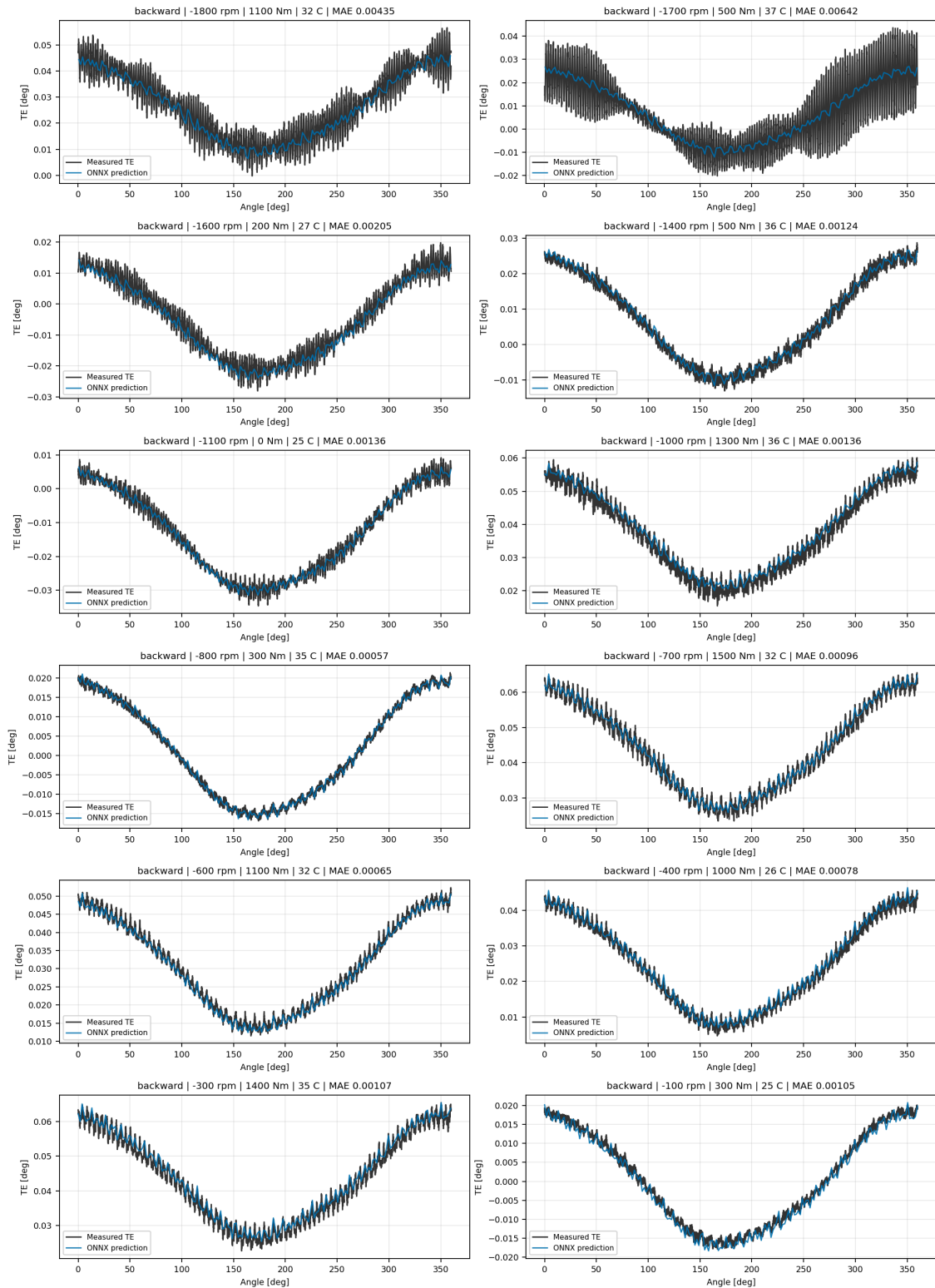
Polished Dataset + Actual Values | wave4_1_smooth_l1_robust_loss | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

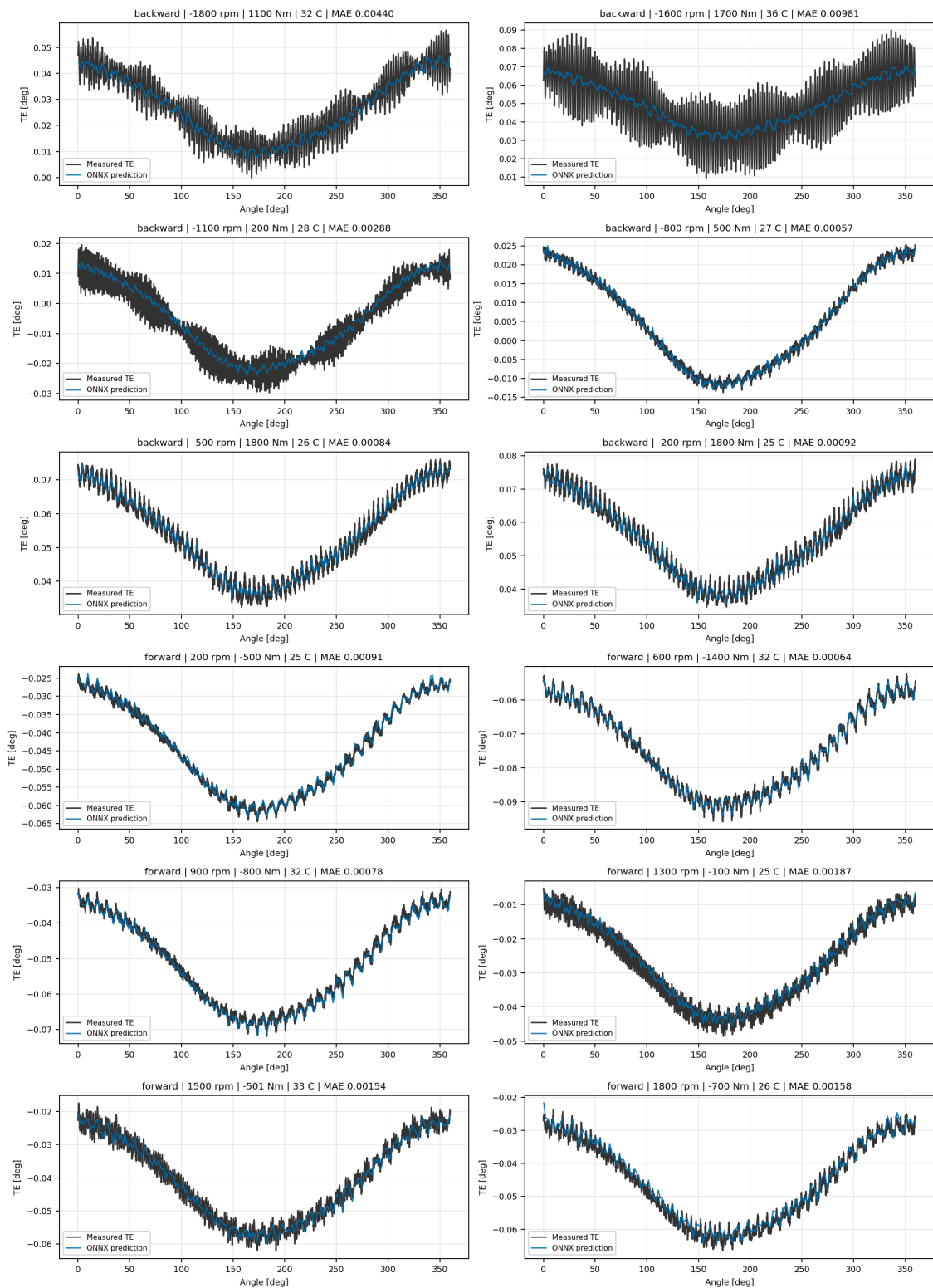
Polished Dataset + Actual Values | wave4_1_smooth_l1_robust_loss | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave4_1_smooth_l1_robust_loss | global



polished_dataset_actual_values global 12-curve collage